Wave guides as an analogous system of 1D optical lattice for anyons

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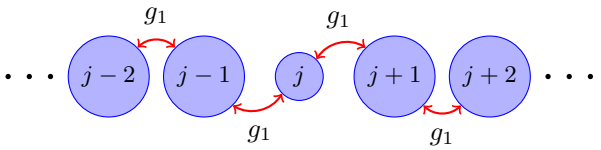
In this work we provide an analytical framework for describing the propagation of a quantum walk along an 1d optical lattice using the analogy of light in waveguide arrays, considering an infinite case for the waveguides, translating to the anyons systems the results from perturbative theory allows to propose an external potential to recreate anyonic behavior in waveguides offering a feasible treatment into the propagation dynamics in these systems, and comparing the result with an non perturbative solution and a numerical solution via exact diagonalization of the Hamiltonian.

Keywords: anyons, quantum matter, ultracold atoms, waveguides

I. INTRODUCTION

Los anyones son ...

II. MAIN IDEA



Para las guías de onda:

- Una guía de onda pose un indice de refracción en el otro núcleo, rodeado por otro de menor indice $n_1 < n_2$
- Para una única guía de onda, los modos admitidos están dados por sus características físicas.
- Un arreglo de guías de onda, el acoplamiento permitirá la transferencia entre los elementos del arreglo.

From an infinite waveguides array with homogeneous optical single mode waveguides and equally spaced along z-axis. This is described by the following differential equation [1]

$$i\frac{dE_j(z)}{dz} = g_1(E_{j-1} + E_{j+1}), \quad j = -\infty, \ldots, \infty, \quad (1)$$

From this

$$i\frac{d|\psi(z)\rangle}{dz} = \hat{H}|\psi(z)\rangle, \quad \hat{H} = g_1\left(\hat{V}_\infty + \hat{V}_\infty^\dagger\right), \quad (2)$$

where the operators $\hat{V}_\infty$ and $\hat{V}_\infty^\dagger$ are the infinite counterparts of the London-Susskind–Glogower operators [2, 3], defined as

$$\hat{V}_\infty = \sum_{n=-\infty}^{\infty} |n\rangle\langle n+1|, \quad \hat{V}_\infty^\dagger = \sum_{n=-\infty}^{\infty} |n+1\rangle\langle n|, \quad (3)$$

For an analytical solution we need to satisfy in our Hilbert space the following conditions for the LSG operator

$$\left[\hat{V}_\infty, \hat{V}_\infty^\dagger\right] = 0, \qquad \hat{V}_\infty\hat{V}_\infty^\dagger = \hat{I},$$

In this framework the initial state is a photon in the n site of an infinite array of waveguides $|\psi(0)\rangle = |n_0\rangle$,:

$$|\psi_{n_0}(z)\rangle = \exp\left[-g_1 z\left(i\hat{V}_\infty^\dagger - \frac{1}{i\hat{V}_\infty^\dagger}\right)\right]|n_0\rangle\,.$$

Donde podemos identificar la función generadora de las funciones Bessel, para obtener

$$|\psi_{n_0}(z)\rangle = \sum_{n=-\infty}^{\infty} J_n(-2g_1 z) i^n \hat{V}_\infty^{\dagger n}|n_0\rangle\,.$$

para este modelo podemos hacer una analogía con un modelo de bosones en una red óptica partiendo del operador $\hat{V}_\infty$ y la evolución será escrita en función del tiempo:

$$\hat{V}_\infty = \sum_{j=-\infty}^{\infty} \hat{b}_j^\dagger\hat{b}_{j+1}, \quad \hat{V}_\infty^\dagger = \sum_{j=-\infty}^{\infty} \hat{b}_{j+1}^\dagger\hat{b}_j, \quad (4)$$

Rewriting the Hamiltonian as

$$\hat{H} = -J\sum_{j=-\infty}^{\infty}\left(\hat{b}_j^\dagger\hat{b}_{j+1} + \hat{b}_{j+1}^\dagger\hat{b}_j\right) \quad (5)$$

The operator $\hat{V}_\infty^{\dagger n}$ is equivalent of create a particle in n site in the framework of optical lattices for a subspace of one particle per site:

$$\hat{V}_\infty^{\dagger n}|1_0\rangle = \hat{b}_n^\dagger|0\rangle, \quad (6)$$

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Obtaining the dynamics for a particle in 0-site

$$|\psi_0(t)\rangle = \sum_{n=-\infty}^{\infty} J_n(-2Jt) i^n \hat{b}_n^\dagger |0\rangle . \tag{7}$$

We could obtain a rich dynamics for n-particles system working with a Heisenberg representation for one body operators $\hat{b}^\dagger$ applied in the sites, obtaining in general the expression:

$$\langle \hat{n}_j \rangle = \sum_{m=-\infty}^{\infty} |J_{m-j}(-2Jt)|^2 \langle \hat{n}_m(0) \rangle \tag{8}$$

$$\langle \mathbf{J}_j \rangle = iJ \left\langle \hat{b}_{j+1}^\dagger \hat{b}_j - \hat{b}_j^\dagger \hat{b}_{j+1} \right\rangle \tag{9}$$

$$\langle \mathbf{T}_j \rangle = \sum_{m=-\infty}^{\infty} |J_{m-j}(-2Jt)|^2 \langle \hat{n}_m(0) \rangle \tag{10}$$

also:

$$\mathbf{J} = iJ \sum_{i=1}^{M} \left(e^{-i\theta \hat{n}_i} \hat{b}_{i+1}^\dagger \hat{b}_i - \hat{b}_i^\dagger \hat{b}_{i+1} e^{i\theta \hat{n}_i} \right) \tag{11}$$

$$\mathbf{T} = J \sum_{i=1}^{M} \left(e^{-i\theta \hat{n}_i} \hat{b}_{i+1}^\dagger \hat{b}_i + \hat{b}_i^\dagger \hat{b}_{i+1} e^{i\theta \hat{n}_i} \right) \tag{12}$$

$$\mathrm{S}(q) = \sum_{l=0}^{M} |\mathrm{J}_l(-2Jt)|^2 - \sum_{l,m=0}^{M} e^{\mathrm{iq}(l-m)} |\mathrm{J}_l(-2Jt)|^2 |\mathrm{J}_m(-2Jt)|^2$$

III. ANYONIC MODEL

We study the 1D Anyon-Hubbard model [4] in the purely kinetic regime ($U = 0$). Using the Fractional Jordan-Wigner (FJW) transformation

$$\hat{a}_m = \hat{b}_m \exp\left(i\theta \sum_{l=1}^{m-1} \hat{n}_l \right), \tag{13}$$

the Hamiltonian maps onto bosons $\hat{b}_m, \hat{b}_m^\dagger$ ($[\hat{b}_j, \hat{b}_l^\dagger] = \delta_{j,l}$, $\hat{n}_m = \hat{b}_m^\dagger \hat{b}_m$) with density-dependent hopping phases:

$$\hat{H}_{+\theta}^b = -J \sum_{m=1}^{M-1} \left(\hat{b}_m^\dagger \hat{b}_{m+1} e^{i\theta \hat{n}_m} + e^{-i\theta \hat{n}_m} \hat{b}_{m+1}^\dagger \hat{b}_m \right) \tag{14}$$

where $\theta \in [-\pi, \pi]$ is the statistical angle ($\theta = 0$ for free bosons, $\theta = \pi$ for pseudofermions), obtaining for our model of 2 particles ($N = 2$) on infinity sites (See Appendix S. 4)

$$\begin{aligned} i\frac{d}{dt}\hat{b}_j(t) + J\left[\hat{b}_{j-1}(t) + \hat{b}_{j+1}(t)\right] = \\ -J\Big[\eta_-(\theta)\hat{b}_{j-1}^\dagger(t)\hat{b}_{j-1}^2(t) \\ +2\eta_+(\theta)\hat{b}_j^\dagger(t)\hat{b}_j(t)\hat{b}_{j+1}(t) \\ +\eta_-(\theta)\hat{b}_{j+1}^\dagger(t)\hat{b}_j^2(t)\Big] \end{aligned} \tag{15}$$

A. Non-perturbative solution

We approach the problem with an exact solution via integral $q_t \equiv 2Jt$ so the linear left-hand side of (15) is inverted by the 1D free boson Bessel Green's function:

$$G_{j-l}(q_t) = i^{|j-l|} J_{|j-l|}(q_t), \tag{16}$$

where $J_n(q_t)$ is the Bessel function of the first kind. Applying Duhamel's principle converts (15) into the exact operator integral equation over $\tau \in [0, q_t]$:

$$\begin{aligned} \hat{b}_j(q_t) = \hat{b}_j^{(0)}(q_t) \\ + \frac{i}{2} \sum_{l=-\infty}^{\infty} \int_0^{q_t} d\tau\, G_{j-l}(q_t - \tau)\Big[\eta_-(\theta)\hat{b}_{l-1}^\dagger(\tau)\hat{b}_{l-1}^2(\tau) \\ + 2\eta_+(\theta)\hat{b}_l^\dagger(\tau)\hat{b}_l(\tau)\hat{b}_{l+1}(\tau) \\ + \eta_-(\theta)\hat{b}_{l+1}^\dagger(\tau)\hat{b}_l^2(\tau)\Big], \end{aligned} \tag{17}$$

where $\hat{b}_j^{(0)}(q_t) = \sum_{m=-\infty}^{\infty} G_{j-m}(q_t)\hat{b}_m(0)$ is the free boson operator.

IV. CONCLUSION

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Appendix: Appendix

In this paper we suggest to calculate the operator from active representations due the feasibility of operations

1. Passive representation of $\hat{n}_j$ operator

For this framework we need to specify the initial state, first, for 2 particles in the sites j, j' respectively, with $j \neq j'$, the state as function of time is

$$|\psi(t)\rangle = e^{-i\mathcal{H}t}|\psi(0)\rangle = e^{-i\mathcal{H}t}\hat{b}_j^\dagger \hat{b}_{j'}^\dagger|0\rangle, \tag{A.1}$$

as the temporal propagator is unitary we use the identity to get the following expressions

$$\begin{aligned} |\psi(t)\rangle &= e^{-i\mathcal{H}t}\hat{b}_j^\dagger \mathcal{I} \hat{b}_{j'}^\dagger \mathcal{I}|0\rangle \\ &= e^{-i\mathcal{H}t}\hat{b}_j^\dagger e^{i\mathcal{H}t} e^{-i\mathcal{H}t}\hat{b}_{j'}^\dagger e^{i\mathcal{H}t} e^{-i\mathcal{H}t}|0\rangle \\ &= e^{-i\mathcal{H}t}\hat{b}_j^\dagger e^{i\mathcal{H}t} e^{-i\mathcal{H}t}\hat{b}_{j'}^\dagger e^{i\mathcal{H}t}|0\rangle, \end{aligned} \tag{A.2}$$

as the bosons obey time invariance $\hat{U}^\dagger \hat{b}^\dagger \hat{U} = \hat{U}\hat{b}^\dagger \hat{U}^\dagger$

$$|\psi(t)\rangle = \hat{b}_j^\dagger \hat{b}_{j'}^\dagger|0\rangle, \tag{A.3}$$

we derive the Heisenberg equation of motion $\hat{b}^\dagger(t)$ from

$$i\frac{d}{dt}\hat{b}_j^\dagger = \left[\hat{b}_j^\dagger, \hat{H}\right], \tag{A.4}$$

doing the commutator with (5) obtain:

$$i\frac{d}{dt}\hat{b}_j^\dagger(t) = J\left(\hat{b}_{j-1}^\dagger + \hat{b}_{j+1}^\dagger\right), \tag{A.5}$$

consistent with the phenomenology of (1), to solve this differential equation we can use the Fourier transform $\hat{b}_j^\dagger = \frac{1}{\sqrt{N}}\sum_k e^{-ikj}\hat{b}_k^\dagger$ and return to the original space with an inverse Fourier transform, so in momentum space the differential equation decouples:

$$i\frac{d}{dt}\hat{b}_k^\dagger(t) = 2J\cos(k)\hat{b}_k^\dagger(t), \tag{A.6}$$

Solving

$$\hat{b}_k^\dagger(t) = e^{-i2Jt\cos(k)}\hat{b}_k^\dagger(0), \tag{A.7}$$

and taking the continuum limit to transform back to time space using the inverse Fourier transform

$$\hat{b}_j^\dagger(t) = \frac{1}{2\pi}\int_{-\pi}^{\pi} e^{-ikj}\left(e^{-i2Jt\cos(k)}\hat{b}_k^\dagger(0)\right)dk, \tag{A.8}$$

substituting $\hat{b}_k^\dagger(0) = \sum_m e^{ikm}\hat{b}_m^\dagger(0)$

$$\hat{b}_j^\dagger(t) = \sum_m \left(\frac{1}{2\pi}\int_{-\pi}^{\pi} e^{-i2Jt\cos(k)}e^{ik(m-j)}\hat{b}_m^\dagger(0)dk\right), \tag{A.9}$$

using the Jacobi Anger expansion:

$$e^{iz\cos(k)} = \sum_{n=-\infty}^{\infty} i^n J_n(z)e^{ink}. \tag{A.10}$$

Substituting with $z = -2Jt$, we obtain

$$\hat{b}_j^\dagger(t) = \sum_{m,n}\left(\frac{1}{2\pi}\int_{-\pi}^{\pi} i^n J_n(-2Jt)e^{ik(m+n-j)}\hat{b}_m^\dagger(0)dk\right), \tag{A.11}$$

due orthogonality of complex exponentials functions, the integral will be zero unless $n + m - j = 0$, so for the values $n = j - m$ and using properties of Bessel function, we get obtain the the expression:

$$\hat{b}_j^\dagger(t) = \sum_{m=-\infty}^{\infty} i^{m-j}J_{m-j}(-2Jt)\hat{b}_m^\dagger(0) \tag{A.12}$$

2. Active representation of $\hat{n}_j$ operator

We obtain for a $|\psi\rangle$ as steady state and $\hat{b}^\dagger$ as dynamic operator, write as:

$$\hat{b}_j^\dagger(t) = \sum_{m=-\infty}^{\infty} i^{m-j}J_{m-j}(-2Jt)\hat{b}_m^\dagger(0), \tag{A.13}$$

so the expected value of $\hat{n}_j$ is:

$$\begin{aligned} \langle\hat{n}_j(t)\rangle &= \langle\psi(0)|\hat{b}_j^\dagger(t)\hat{b}_j(t)|\psi(0)\rangle, \\ &= \langle\psi(0)|\sum_{m,m'=-\infty}^{\infty} i^{m-m'}J_{m-j}(-2Jt)J_{m'-j}(-2Jt) \\ &\quad \hat{b}_m^\dagger\hat{b}_m|\psi(0)\rangle, \end{aligned} \tag{A.14}$$

In general we get a 1d chain, if we annihilate at m' site and create at m we will obtain an orthogonal state, so for the different values of $m \neq m'$ we will obtain zero, so for an arbitrary initial state we get:

$$\langle\hat{n}_j(t)\rangle = \sum_{m=-\infty}^{\infty}|J_{m-j}(2Jt)|^2\langle\hat{n}_m(0)\rangle \tag{A.15}$$

3. Dynamical properties

In general for the active operators are defined with the temporal propagator

$$\langle \hat{O}(t)\rangle_{+\theta} = \langle \Psi_0 | e^{i\hat{H}_{B,+\theta}t}\hat{O}e^{-i\hat{H}_{B,+\theta}t}|\Psi_0\rangle \tag{A.16}$$

and it is useful to write the $\mathcal{K}$ invariance defined in [5] to obtain some dynamical properties of the observable:

$$\begin{aligned}\langle \hat{O}(t)\rangle_{+\theta} = \langle \Psi_0|\mathcal{K}^\dagger \left(\mathcal{K}e^{i\hat{H}_{B,+\theta}t}\mathcal{K}^\dagger\right) \\ \left(\mathcal{K}\hat{O}\mathcal{K}^\dagger\right)\left(\mathcal{K}e^{-i\hat{H}_{B,+\theta}t}\mathcal{K}^\dagger\right) \\ \left(\mathcal{K}e^{-i\hat{H}_{B,+\theta}t}\mathcal{K}^\dagger\right)\mathcal{K}|\Psi_0\rangle\end{aligned} \tag{A.17}$$

One should pair $\mathcal{K}$ with $\mathcal{T}$ or $\mathcal{P}$ to restores forward time evolution at angle $-\theta$ or $-U$ respectively, obtaining the following dynamical properties, via $\mathcal{TK}$ ($\theta \leftrightarrow -\theta$):

$$\langle \hat{n}_j(t)\rangle_{+\theta} = \langle \hat{n}_{j'}(t)\rangle_{-\theta}, \tag{A.18}$$

$$\langle \hat{K}^{\text{anyon}}_{j,j+1}(t)\rangle_{+\theta} = \langle \hat{K}^{\text{anyon}}_{j'-1,j'}(t)\rangle_{-\theta}, \tag{A.19}$$

$$\langle \hat{I}^{\text{anyon}}_{j,j+1}(t)\rangle_{+\theta} = -\langle \hat{I}^{\text{anyon}}_{j'-1,j'}(t)\rangle_{-\theta}, \tag{A.20}$$

and via $\mathcal{PK}$ ($U \leftrightarrow -U$)

$$\langle \hat{n}_j(t)\rangle_{+U} = \langle \hat{n}_{j'}(t)\rangle_{-U}, \tag{A.21}$$

$$\langle \hat{K}^{\text{anyon}}_{j,j+1}(t)\rangle_{+U} = -\langle \hat{K}^{\text{anyon}}_{j'-1,j'}(t)\rangle_{-U}, \tag{A.22}$$

$$\langle \hat{I}^{\text{anyon}}_{j,j+1}(t)\rangle_{+U} = -\langle \hat{I}^{\text{anyon}}_{j'-1,j'}(t)\rangle_{-U}, \tag{A.23}$$

giving us an insight about the symmetry or the asymmetry of the quantum walk.

4. Solving the differential equation for the anyonic system

To derive the Heisenberg equation of motion,

$$i\frac{d}{dt}\hat{b}_j(t) = [\hat{b}_j(t), \hat{H}^b_{+\theta}], \tag{A.24}$$

In this case we can use the following bosonic identities:

$$\hat{b}_j e^{\pm i\theta\hat{n}_j} = e^{\pm i\theta(\hat{n}_j+1)}\hat{b}_j = e^{\pm i\theta}e^{\pm i\theta\hat{n}_j}\hat{b}_j, \tag{A.25}$$

$$\hat{b}^\dagger_j e^{\pm i\theta\hat{n}_j} = e^{\mp i\theta}e^{\pm i\theta\hat{n}_j}\hat{b}^\dagger_j, \tag{A.26}$$

Only the adjacent sites $m = j-1$ and $m = j$ have non-zero commutators with $\hat{b}_j$, yielding the general operator differential equation:

$$\begin{aligned}i\frac{d}{dt}\hat{b}_j = -J\Big\{ & e^{-i\theta\hat{n}_{j-1}}\hat{b}_{j-1} \\ & + e^{i\theta\hat{n}_j}\left[1+\left(1-e^{-i\theta}\right)\hat{n}_j\right]\hat{b}_{j+1} \\ & + \left(e^{-i\theta}-1\right)\hat{b}^\dagger_{j+1}e^{-i\theta\hat{n}_j}\hat{b}^2_j\Big\}\end{aligned} \tag{A.27}$$

To work with this equation we need to simplify some exponential terms to polynomial terms with a truncation, defining a maximum value for particles in the chain. When the total particle number in the optical lattice is fixed to $N = 2$, the local occupation number in all sites never exceeds $n_{\max} = 2$. Acting (A.27) on any two-particle state $|\Psi(0)\rangle$ in $H^b_{N=2}$ restricts the occupation numbers seen by the exponential phase operators $e^{\pm i\theta\hat{n}}$. In the third term, $\hat{b}^2_j$ acts first and removes both particles from site j, leaving only the $n_j = 0$ state behind. Thus $e^{-i\theta\hat{n}_j}\hat{b}^2_j = e^0\hat{b}^2_j = \hat{b}^2_j$. The first and second term got a single annihilation ($\hat{b}_{j-1}$ and $\hat{b}_{j+1}$), once we apply $\hat{b}_{j-1}$ or $\hat{b}_{j+1}$ removes 1 particle, there is at most 1 particle remaining anywhere on the lattice ($n \in \{0,1\}$). On the subspace $\{0,1\}$, any function of $\hat{n}$ is exact at linear order ($e^{\alpha\hat{n}} = 1 + (e^\alpha - 1)\hat{n}$ and $\hat{n}^2 = \hat{n}$), simplifying

$$e^{-i\theta\hat{n}_{j-1}}\hat{b}_{j-1} = \left[1+\left(e^{-i\theta}-1\right)\hat{n}_{j-1}\right]\hat{b}_{j-1} \tag{A.28}$$

$$e^{i\theta\hat{n}_j}\left[1+(1-e^{-i\theta})\hat{n}_j\right]\hat{b}_{j+1} = \left[1+2\left(e^{i\theta}-1\right)\hat{n}_j\right]\hat{b}_{j+1} \tag{A.29}$$

Defining $\eta_\pm(\theta) \equiv e^{\pm i\theta} - 1$ and writing $\hat{n}_j = \hat{b}^\dagger_j\hat{b}_j$, the general differential equation (A.27) reduces to a cubic polynomial differential equation:

$$\begin{aligned}i\frac{d}{dt}\hat{b}_j(t) = & -J\left(\hat{b}_{j-1}+\hat{b}_{j+1}\right) \\ & - J\Big[\left(e^{-i\theta}-1\right)\hat{n}_{j-1}\hat{b}_{j-1} \\ & + 2\left(e^{i\theta}-1\right)\hat{n}_j\hat{b}_{j+1} \\ & + \left(e^{-i\theta}-1\right)\hat{b}_{j+1}\hat{b}^2_j\Big]\end{aligned} \tag{A.30}$$